Shark Colony Experiment

Shared-ocean ecology (GA + RL) — 2026-06-12T08:56:02

Summary

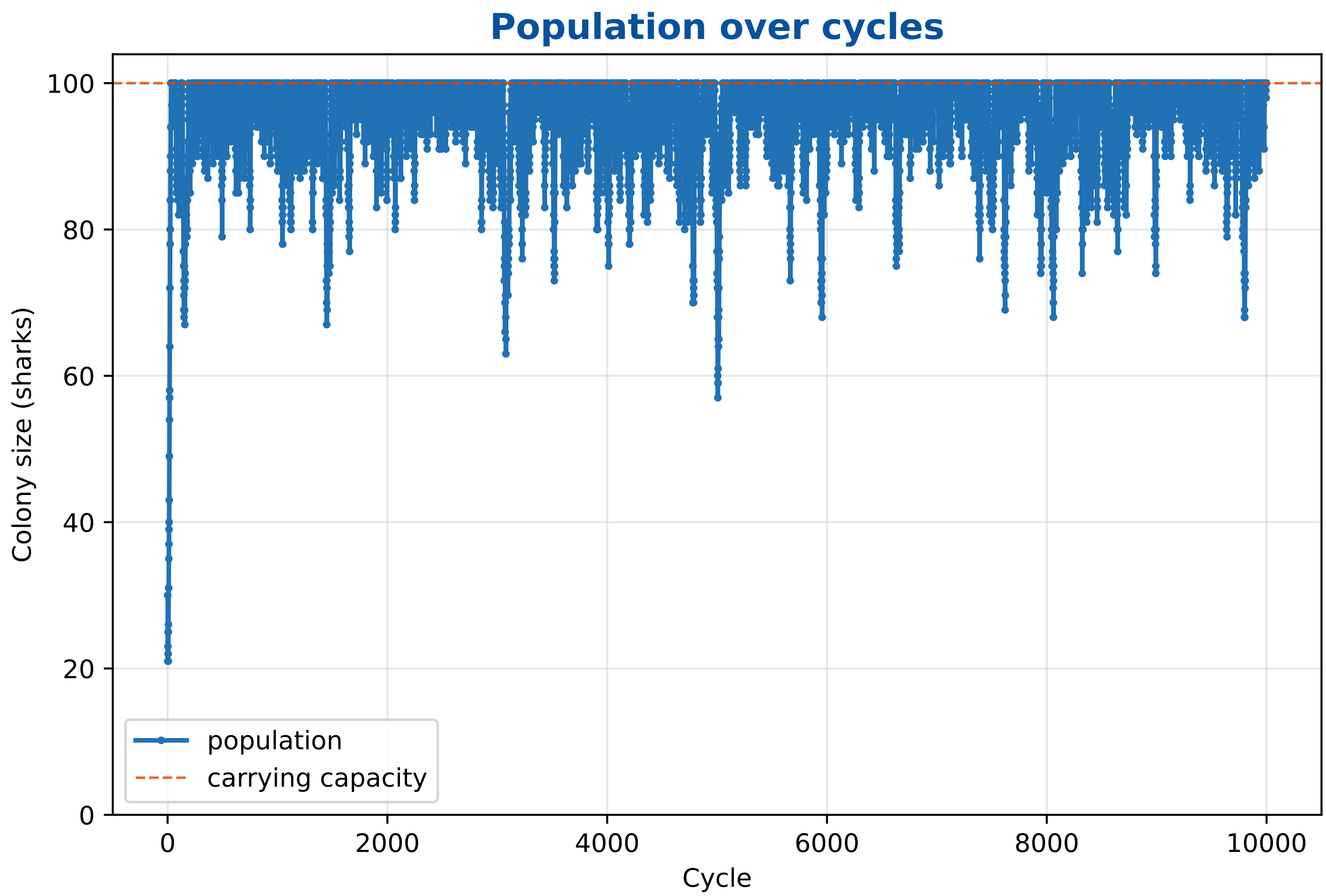
Summary	
Cycles requested / completed	10000 / 10000
Founding pod	30
Carrying capacity	100
Peak population	100
Final population	100
Colony went extinct?	no
Best fitness (any cycle)	278.200
Final mean fitness	-10.721
Runtime	536.6 s

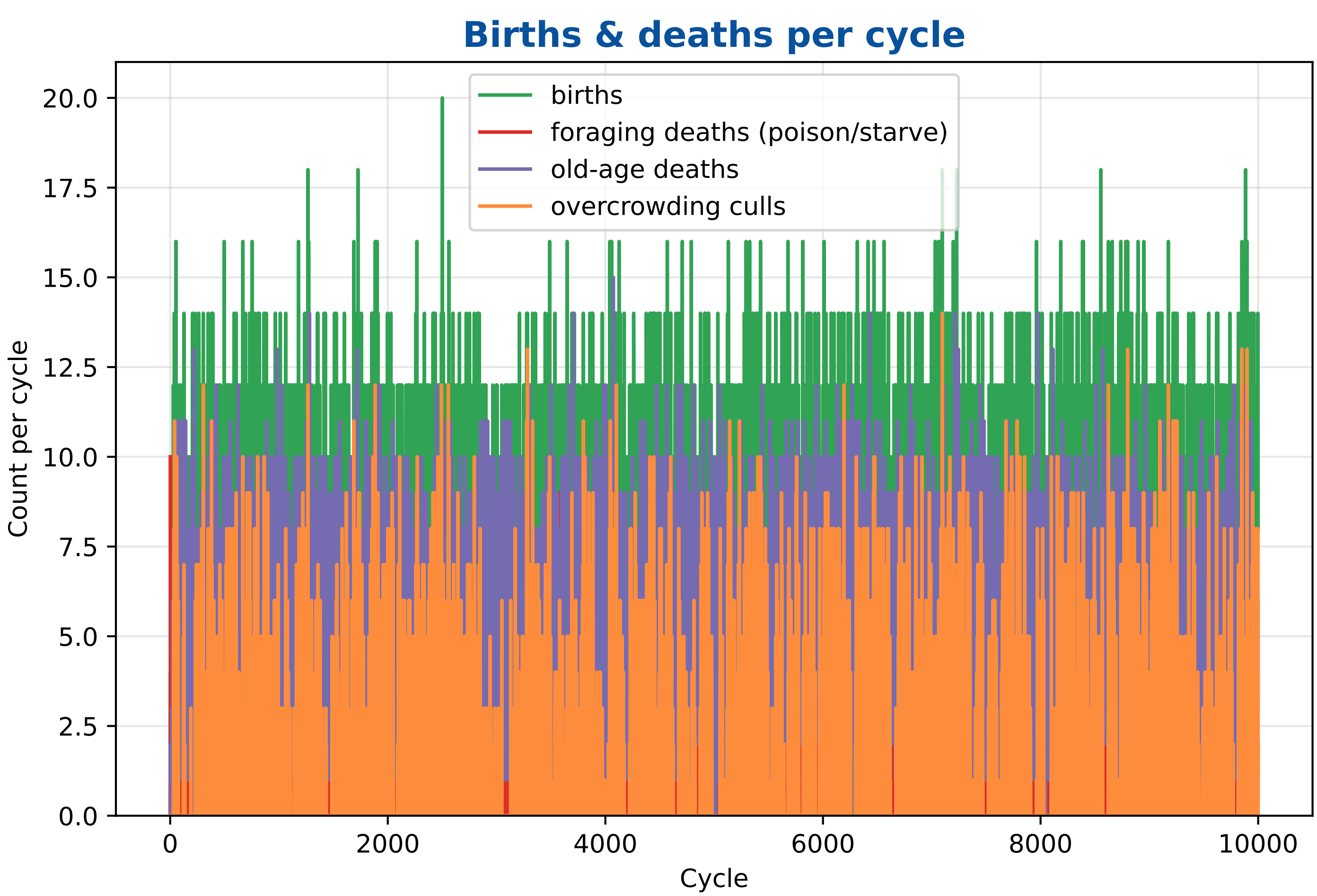
Ecology & competition

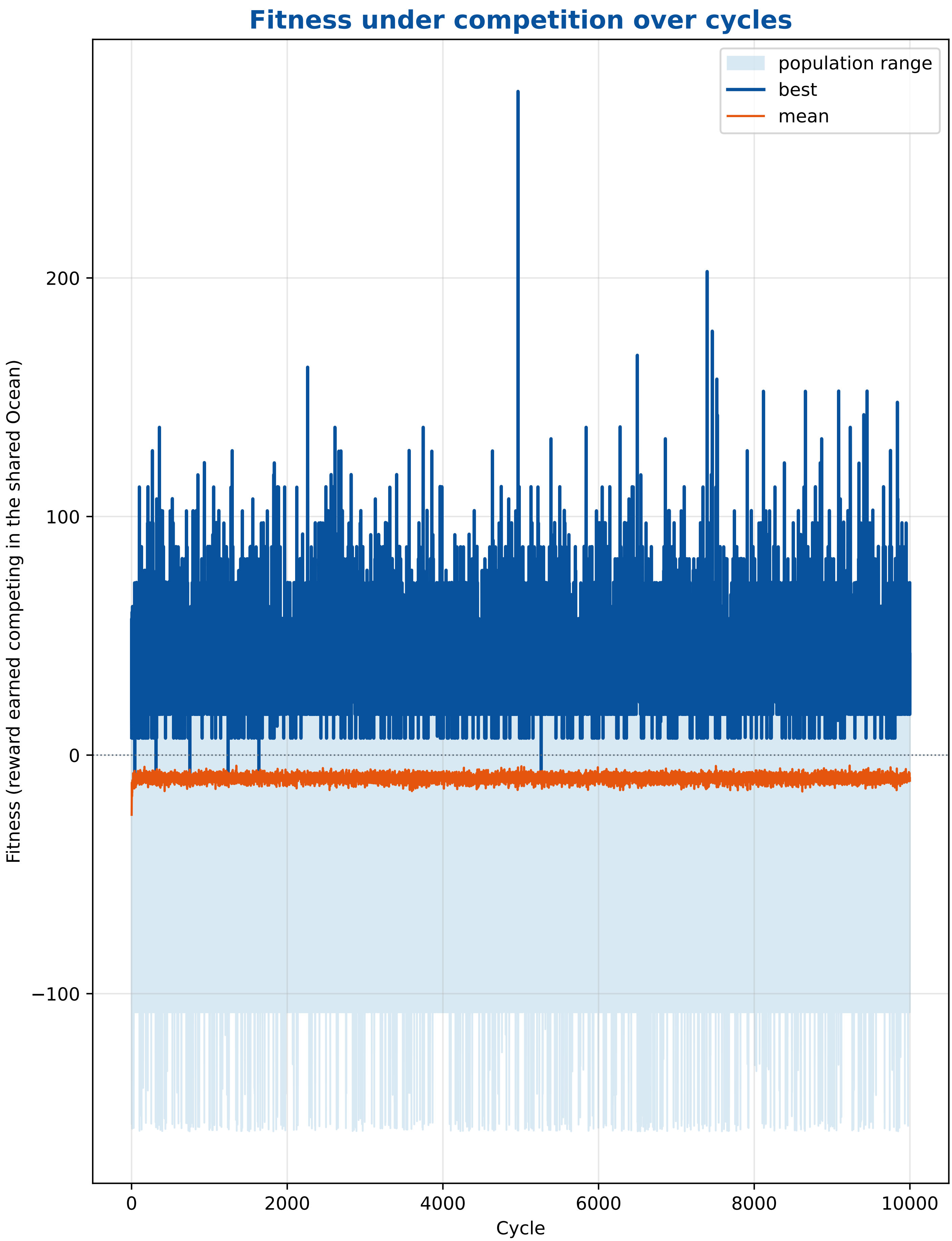
Ecology setting	Value
Shared, depleting ocean	True
Fish per shark / pool cap	1.8 / 30
Must eat to breed	True
Fitness-weighted breeding	True
Cull weakest on overcrowding	True
Gestation divisor / brood	6.0 / 2
Steps per life / brain warmup	80 / 400
Mutation rate / seed	0.2 / 0
Run timestamp	2026-06-12T08:56:02

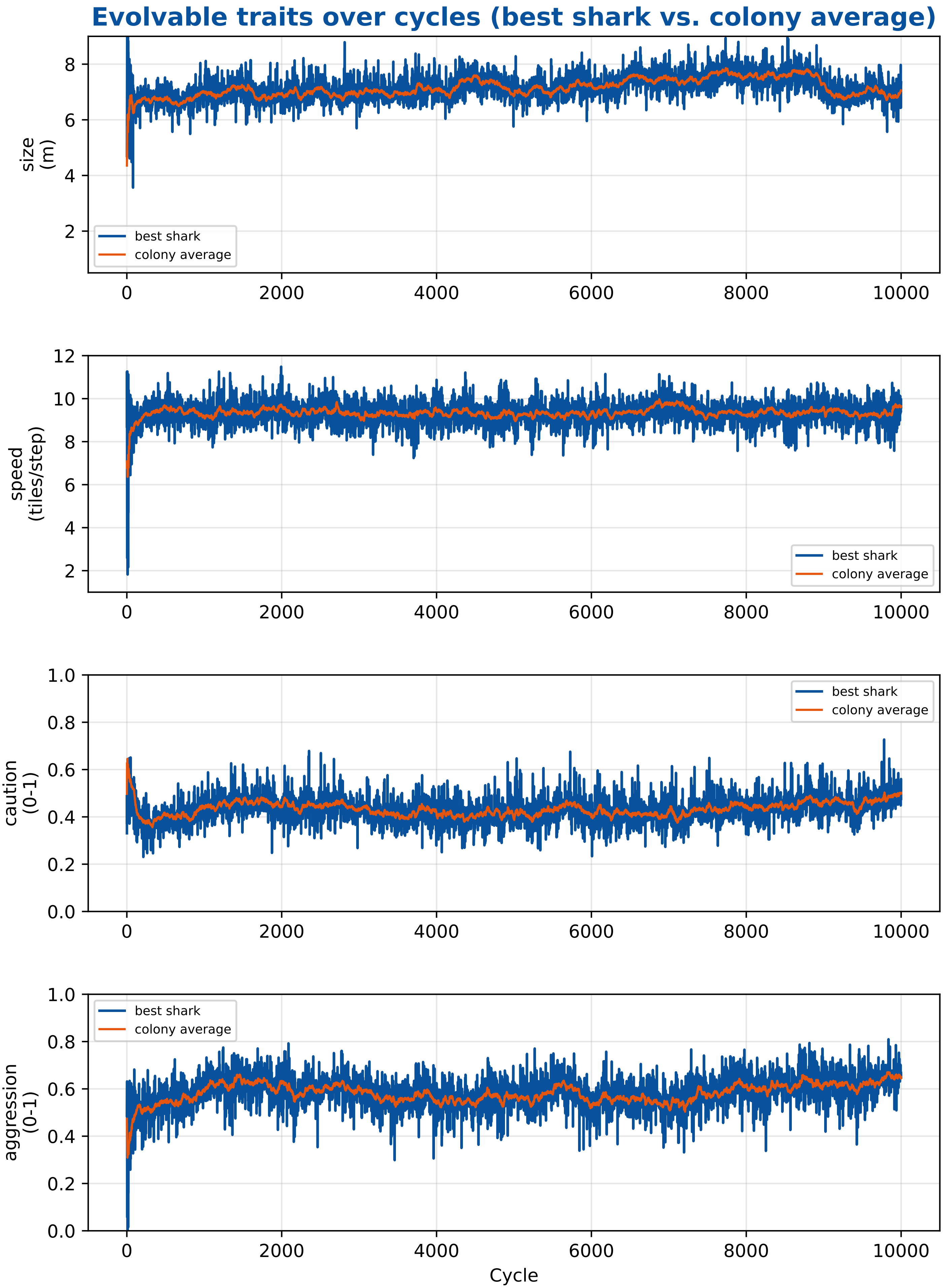
Shared brain (Q-learning)

Shared brain (Q-learning)	Value
States learned	237
Final epsilon (explore rate)	0.050
Epsilon min / decay	0.050 / 0.9950
Alpha (learning rate)	0.100
Gamma (discount)	0.950

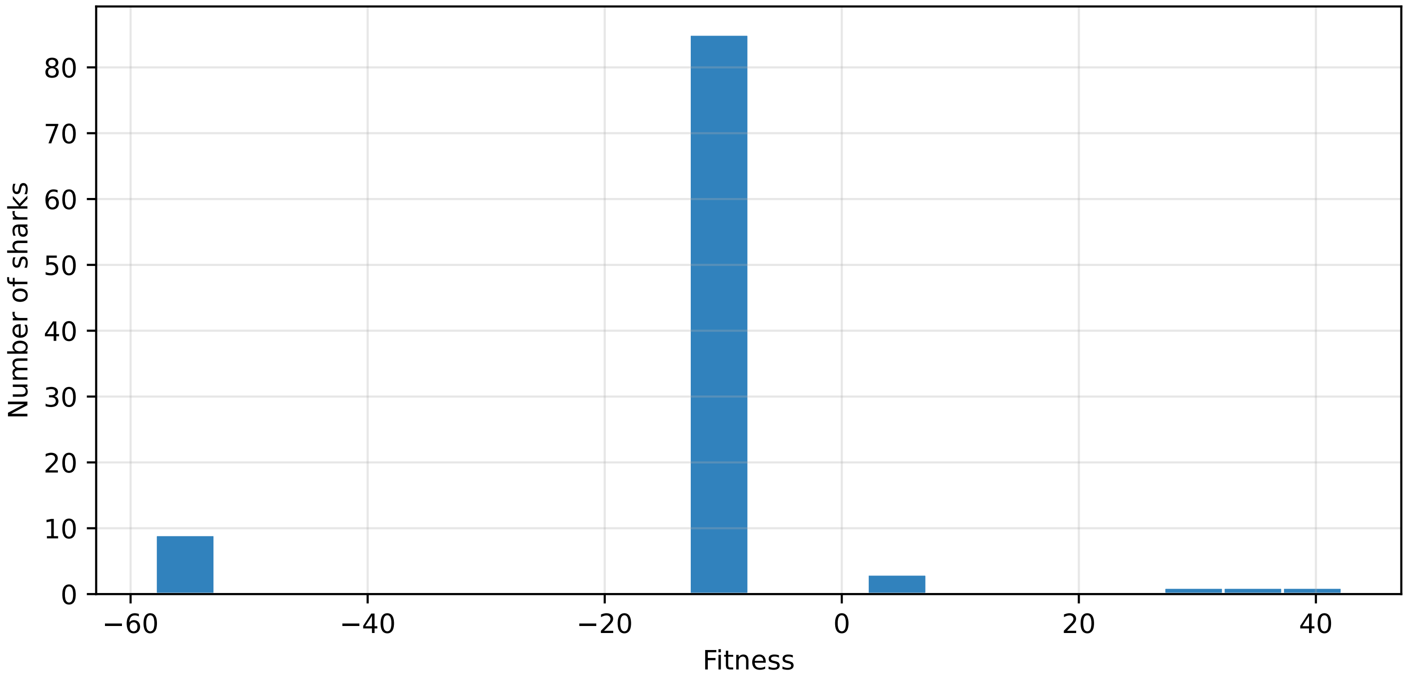

Population over cycles
Colony size (sharks)
Cycle
population
carrying capacity
0
20
40
60
80
100
0
2000
4000
6000
8000
10000


Births & deaths per cycle
Count per cycle
Cycle
births
foraging deaths (poison/starve)
old-age deaths
overcrowding culls
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
2000
4000
6000
8000
10000


Fitness under competition over cycles
population range
best
mean
Fitness (reward earned competing in the shared Ocean)
200
100
0
−100
0
2000
4000
6000
8000
10000
Cycle


Evolvable traits over cycles (best shark vs. colony average)
size
(m)
best shark
colony average
speed
(tiles/step)
caution
(0-1)
aggression
(0-1)
Cycle
0
2000
4000
6000
8000
10000

Final-cycle fitness distribution (cycle 10000)



Best genome found (highest single-cycle reward under competition)

Trait	Value	Unit	Evolved?
size	6.799	m	yes
speed	9.465	tiles/step	yes
caution	0.376	0-1	yes
gestation_period	12.000	months	no (fixed)
aggression	0.521	0-1	yes
field_of_perception	5.000	tiles (radius)	no (fixed)
color_hue	205.000	degrees (HSV hue)	no (fixed)

Per-cycle summary (showing 31 of 10000 cycles, every 334)

Cycle	Pop	Size	Speed	Caution	Aggressior	Best	Mean	Born	Died	Cull
1	30	4.70	8.70	0.33	0.49	7.4	-25.0	2	11	0
335	93	6.80	9.69	0.36	0.53	32.1	-12.5	6	9	0
669	100	6.61	9.99	0.31	0.46	32.2	-10.9	6	4	2
1003	90	7.07	9.18	0.38	0.61	32.2	-8.3	8	5	0
1337	98	7.02	9.31	0.41	0.51	62.3	-6.5	4	5	0
1671	100	6.20	9.53	0.46	0.62	22.2	-11.9	10	9	1
2005	100	6.96	10.33	0.41	0.66	32.2	-7.4	8	6	2
2339	97	6.94	8.96	0.47	0.70	42.2	-12.0	8	8	0
2673	100	6.91	9.26	0.43	0.60	42.2	-8.1	10	4	6
3007	94	6.11	9.28	0.41	0.59	32.2	-9.9	6	6	0
3341	97	6.90	8.82	0.40	0.55	32.2	-9.0	8	6	0
3675	100	6.95	8.99	0.39	0.54	32.1	-6.8	8	7	1
4009	84	7.32	9.03	0.45	0.54	47.2	-8.2	4	3	0
4343	87	7.23	10.53	0.35	0.56	32.1	-13.2	8	4	0
4677	100	7.42	8.61	0.40	0.60	57.3	-9.1	8	5	3
5011	65	7.01	8.94	0.38	0.56	32.1	-11.4	6	6	0
5345	98	7.00	9.00	0.42	0.58	32.1	-10.5	8	2	4
5679	100	6.66	8.52	0.46	0.64	32.2	-9.1	6	9	0
6013	100	7.10	9.36	0.40	0.54	47.2	-8.4	6	10	0
6347	100	6.94	9.25	0.47	0.56	42.2	-9.2	10	8	2
6681	97	7.47	9.17	0.41	0.57	47.2	-8.2	6	4	0
7015	90	7.64	9.98	0.39	0.56	42.2	-10.9	6	7	0
7349	99	7.46	9.45	0.47	0.58	22.2	-9.7	8	6	1
7683	100	8.42	9.61	0.41	0.62	32.2	-9.1	8	9	0
8017	99	7.51	8.62	0.44	0.68	22.2	-11.0	6	8	0
8351	98	7.76	9.18	0.44	0.60	17.1	-13.2	4	6	0
8685	90	7.85	8.67	0.48	0.64	47.2	-8.4	10	8	0
9019	100	7.61	9.46	0.49	0.66	7.1	-11.8	4	8	0
9353	100	6.85	9.21	0.55	0.61	47.2	-7.2	6	8	0
9687	96	6.91	9.84	0.47	0.65	57.2	-7.8	10	7	0
10000	100	7.35	9.60	0.48	0.65	42.2	-10.7	8	8	0